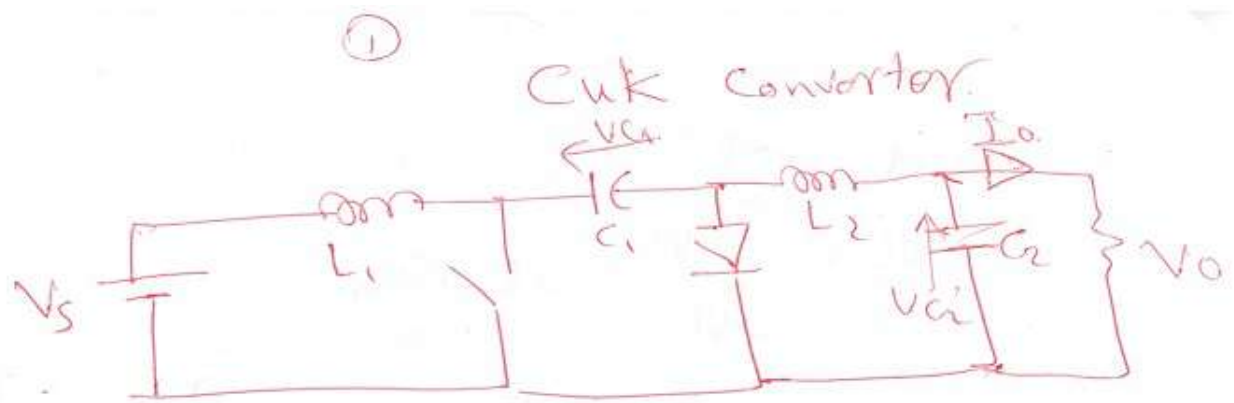
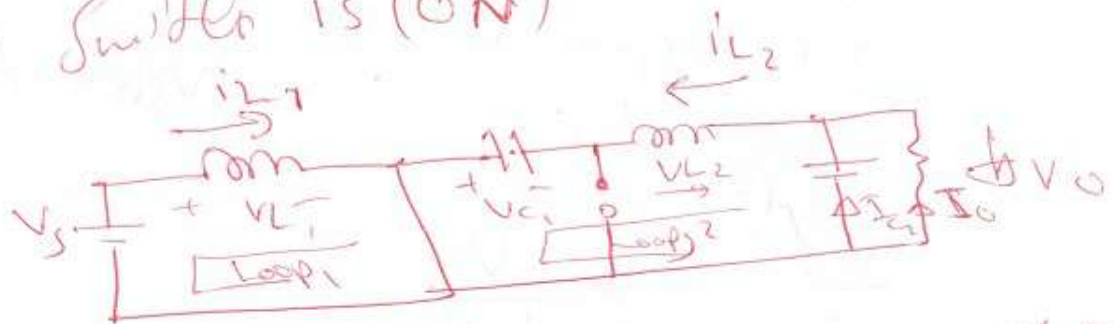




* negative Polarity.

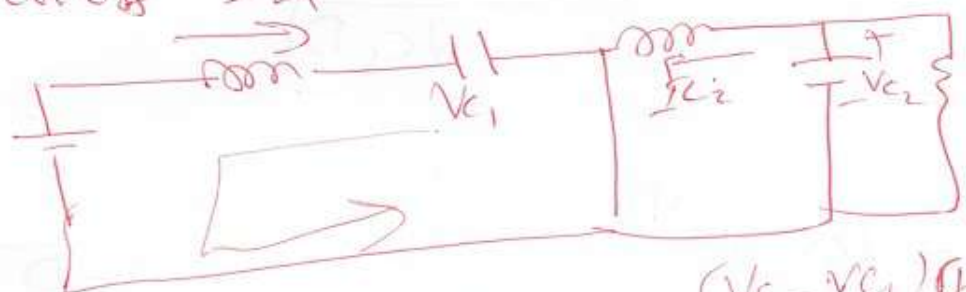
* When Switch is (ON)



1st Loop $V_s = V_{L1} \Rightarrow L_1 \frac{di}{dt} = V_s \Rightarrow \Delta I_{L1} = \frac{V_s DT}{L_1}$ (1)

2nd Loop $V_{C1} - V_{L2} - V_o = 0$
 $V_{L2} = V_{C1} - V_o \Rightarrow L_2 \frac{\Delta I_{L2}}{DT} = (V_{C1} - V_o)$
 $\therefore \Delta I_{L2} = \frac{(V_{C1} - V_o) DT}{L_2}$ (2)

When Switch off I_{L1}



$V_s - V_{C1} = V_{L1} \Rightarrow \Delta I_{L1} = \frac{(V_s - V_{C1})(1-D)T}{L_1}$

②

Proof Loop 2

$$V_{L2} = \oplus V_O$$

$$V_O = \oplus V_{L2} = \frac{\oplus L_1 \Delta I_{L1}}{(1-D)T}$$

$$\therefore \Delta I_{L2 \text{ open}} = \frac{\oplus V_O (1-D)T}{L_2}$$

$$\Delta I_{L1} + \Delta I_{L2} = 0$$

$$\frac{V_S D T}{L_1} = \frac{(V_{C1} - V_S) (1-D) T}{L_1} \Rightarrow V_{SD} = V_{C1} - V_{C1} D$$

$$\Rightarrow V_{C1} = \frac{V_S}{1-D}$$

by equating loops (2)

$$\frac{(V_{C1} - V_O) D T}{L_2} = \frac{V_O (1-D) T}{L_2}$$

$$V_{C1} D - V_O D = V_O - V_O D$$

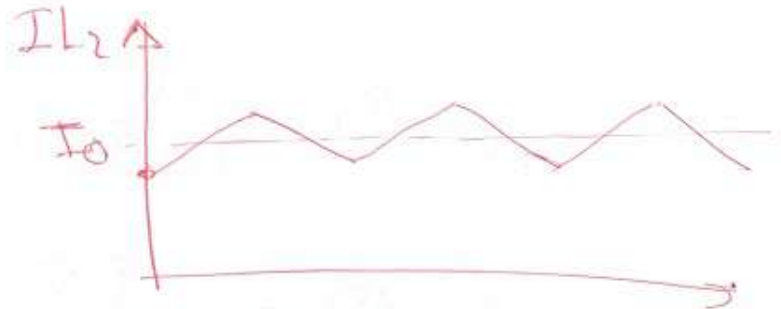
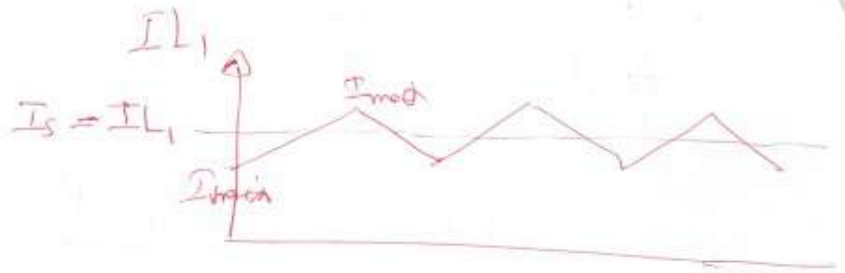
$$\boxed{\therefore V_O = V_{C1} D}$$

$$\therefore V_{C1} = \frac{V_S}{1-D} \Rightarrow V_O = \frac{D V_S}{1-D}$$

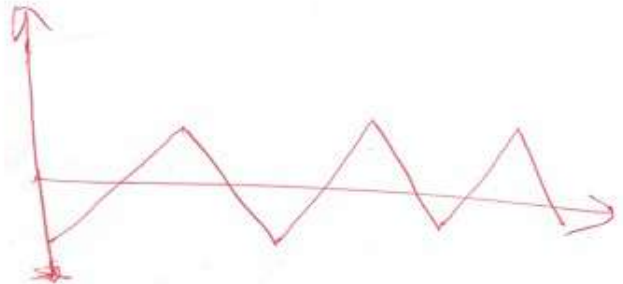
$$\boxed{\therefore \frac{V_O}{V_S} = \frac{D}{1-D}}$$

Curves:-

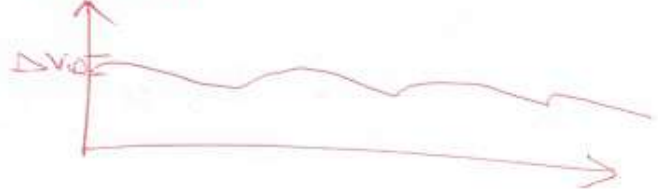
②



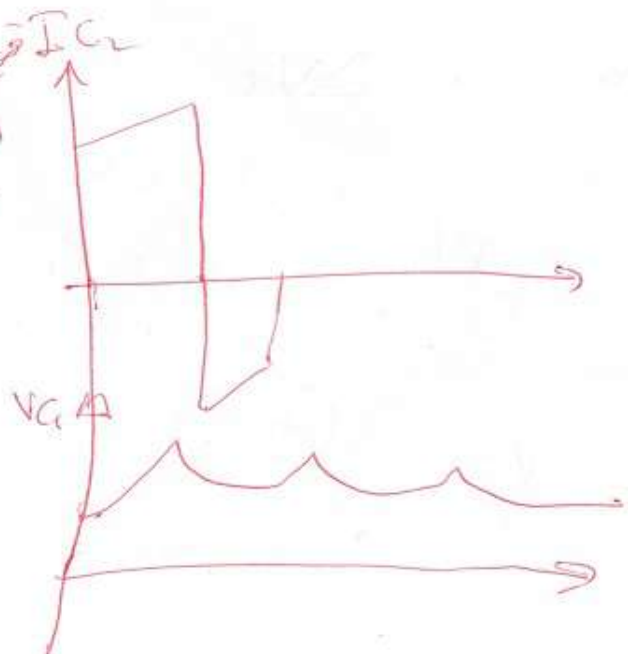
$$I_{C2} = I_{L2} - I_o$$



$$V_o = V_{C2}$$



التيار في حثية I_{C2} افضل من التيار في حثية I_{L2} لان التيار في حثية I_{L2} يكون موجبا دائما والتيار في حثية I_{C2} يكون موجبا وسلبا



(4)

to find ΔIL_2 from Loop 2

$$\Delta IL_2 = \frac{(V_G - V_O) D}{f L_2}$$

$$\therefore V_G = \frac{V_S}{1-D}, \quad V_O = \frac{D V_S}{1-D}$$

$$\Rightarrow \boxed{\Delta IL_2 = \frac{D V_S}{f L_2}}$$

$$V_S I_S = V_O I_O$$

$$V_S I_S = \frac{D V_S}{1-D} I_O$$

$$\Rightarrow \boxed{I_S = \frac{D}{1-D} I_O}$$

to find ΔV_C

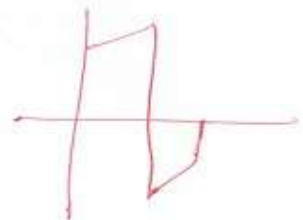
$$Q = C V$$

$$\Delta Q = C \Delta V$$

$$I_C = C \frac{dV}{dt}$$

$$\therefore \Delta V = \frac{1}{C} \int I_C dt = \frac{I_O D T}{C}$$

$$\Delta V_C = \frac{\frac{D I_O}{f C_1}}{RC_1} = \frac{D V_O}{RC_1 f}$$



⑤

to find $\Delta V_{C2} :-$

Buck converter

$$\Delta V_{C2} = \frac{1}{C} \int I_{C2} dt$$

$$\Delta V_{C2} = \frac{1}{2} \times \frac{1}{C} \times \frac{T}{2} \times \frac{\Delta I_{L2}}{2}$$

$$\boxed{\Delta V_o} = \Delta V_{C1} = \frac{\Delta V_s}{8 f^2 L_2 C_2}$$

$$\Delta I_{L1} \min = I_s - \frac{\Delta I_{L1}}{2}$$

$$\Delta I_{L1} = \frac{V_s DT}{L_1}$$

$$I_s = \frac{D}{1-D} I_o$$

$$\therefore \Delta I_{L1} \min = 0$$

$$\frac{D}{1-D} I_o = \frac{V_s DT}{2 L_1}$$

$$\frac{D V_o}{R(1-D)} = \frac{V_s DT}{2 L_1}$$

$$\frac{D V_s}{R(1-D)^2} = \frac{V_s}{2 L_1 f}$$

$$\therefore L_{1 \min} = \frac{(1-D)^2 R}{2 D f}$$

(6)

$$\Delta I L_{2\min} = I_0 - \frac{\Delta I L_2}{2}$$

$$\Delta I L_2 = \frac{V_0(1-D)}{L_2 f} = \frac{DV_S}{L_2 f}$$

$$\therefore \Delta I L_{2\min} = 0 = \frac{V_0}{R} - \frac{DV_S}{2L_2 f}$$

$$\frac{V_0}{(1-D)R} = \frac{DV_S}{2L_2 f}$$

$$2L_2 f (1-D)R$$

$$\therefore L_{2\min} = \frac{(1-D)R}{2f}$$

2. Q1 in the buck converter circuit shown the operating freq is 50 KHz. if the duty ratio is 0.6 and the load current $I_o = 5A$. Assuming Ideal Component calculate

① the average value of V_{C1}

Solution

① from external loop

$$V_s - V_{L1} - V_{C1} + V_{L2} + V_o = 0$$

The average value of the Inductor current is zero for steady state condition

$$V_{L1} = V_{L2} = 0$$

$$\therefore -V_{C1} = V_s + V_o$$

$$V_o = V_s \frac{D}{1-D} = 20 \times \frac{0.6}{1-0.6} = 30V$$

$$\therefore V_{C1} = 20 + 30 = 50V, R = \frac{V_o}{I_o}$$

also determine I_{L1} (max and min)
 I_{L2} (max and min)

$$I_s V_s \geq I_o V_o \\ \therefore I_s \geq 7.5$$

Q2 The input voltage of a Cuk converter is 12 V
 the duty ratio is 0.25 and the switching
 frequency is 25 kHz, $L_1 = 180 \mu H$
 $C_1 = 200 \mu F$, $C_2 = 220 \mu F$ and the average
 output current = 1.25 A

Find

- ① Average output voltage
- ② Average Input Current
- ③ p-p Ripple Current of L_1
- ④ p-p Ripple Voltage of capacitor C_2
- ⑤ p-p Ripple Current of Inductor L_2

Sol

$$\textcircled{1} V_o = \frac{D}{1-D} V_s = \frac{0.25}{1-0.25} \times 12 = 4V$$

$\textcircled{2} I_{L_1} = I_s$ average Inductor current

$$V_s I_s = V_o I_o$$

$$\therefore I_s = \frac{V_o I_o}{V_s} = \frac{4 \times 1.25}{12} = 0.42 A$$

③

$$\textcircled{3} \Delta I_{L1} = \frac{DV_s}{fL_1} = 0.667A$$

$$\textcircled{4} DV_G \approx \frac{DV_o}{R_G f} = 63mV$$

$$\textcircled{5} \Delta I_{L2} = \frac{V_o(1-D)}{L_2 f} \approx 0.8A$$